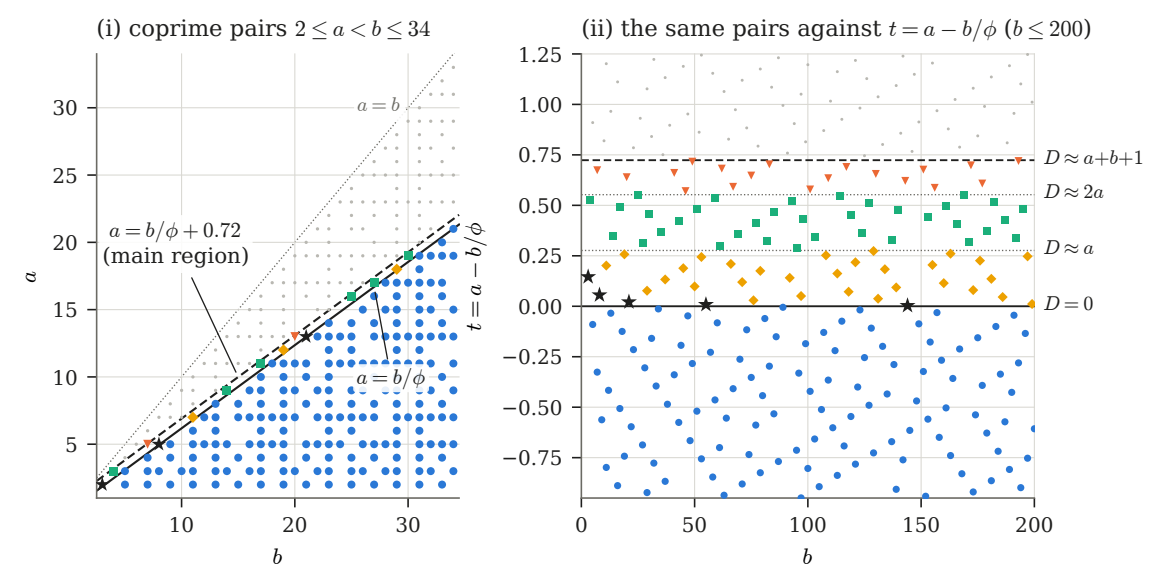
(i) coprime pairs $2 \le a < b \le 34$
(ii) the same pairs against $t = a - b/\phi$ ($b \le 200$)
$a = b$
$a = b/\phi + 0.72$
(main region)
$a = b/\phi$
$t = a - b/\phi$
a
b
$D \approx a+b+1$
$D \approx 2a$
$D \approx a$
$D = 0$
$D \le 0$: all k (golden-ratio region)
$D = 1$: all k (consecutive Fibonacci)
$a+1 \le D \le 2a$: all k
$2 \le D \le a$: fails first at $k = 5$
$2a+1 \le D \le a+b+1$: fails first at $k = 4$
$D > a+b+1$: outside the main region